

Addressing

Chapter 3

Department of Computer Science & Informatics



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Module Objectives

By the end of this module, you should be able to:

1. Work with MAC addresses
2. Configure TCP/IP settings on a computer, including IP address, subnet mask, default gateway, and DNS servers
3. Identify the ports of several common network protocols
4. Describe domain names and the name resolution process
5. Use command-line tools to troubleshoot common network problems

Addressing Overview

- There are four addressing methods:
 - Data link layer MAC (Media Access Control) address is 48 bits, written as six pairs of hex numbers separated by colons
 - It is also called a physical address
 - MAC addresses are embedded on every NIC in the world
 - Network layer IP (Internet Address) address can be used to find any computer in the world
 - IPv4 addresses have 32 bits and are written as four decimal numbers called octets
 - IPv6 addresses have 128 bits and are written as eight blocks of hexadecimal number
 - Transport layer port numbers are used to find applications
 - Application layer FQDNs, computer names, and host names
 - Fully qualified domain name (FQDN) – a unique character-based name

MAC Addresses (1 of 2)

- Traditional MAC addresses contain two parts
 - The first 24 bits are known as the **OUI (Organizationally Unique Identifier)** or manufacturer-ID
 - This part is assigned by the IEEE
 - The last 24 bits make up the **extension identifier** or **device ID**
 - Manufacturer's assign each NIC a unique device ID
- Switches use MAC addresses to identify devices on the local area network

MAC Addresses (2 of 2)

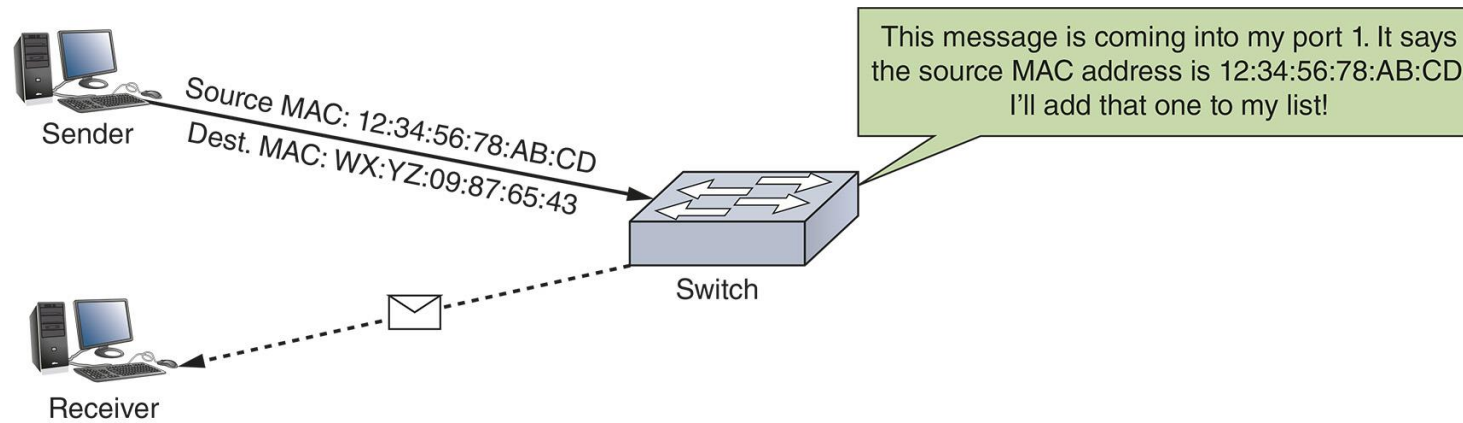


Figure 3-3 The switch learns the sending device's MAC address

Figure 3-3 The switch learns the sending device's MAC address

IP Addresses

- Static IP addresses are assigned manually by the network administrator
- Dynamic IP addresses are automatically assigned by a DHCP server
 - You'll learn more about DHCP later in the chapter
- There are two types of IP addresses:
 - IPv4 is a 32-bit address
 - IPv6 is a 128-bit address

IPv4 Addresses (1 of 5)

- A 32-bit IPv4 address is organized into four groups of 8 bits each (known as octets)
 - Each of the four octets can be any number from 0 to 255
 - Some IP addresses are reserved
- Example of an IPv4 address: 72.56.105.12
- Classful addressing
 - The dividing line between the network and host portions is determined by the numerical range the IP address falls in
- Classful IPv4 addresses are divided into five classes:
 - Class A, Class B, Class C, Class D, and Class E

IPv4 Addresses (2 of 5)

Five Different Classes of IPv4 Addresses

Class	First Octet decimal (range)	First Octet binary (range)	IP range	Subnet Mask	Hosts per Network ID	# of networks
Class A	0 – 127	0XXXXXXXX	0.0.0.0-127.255.255.255	255.0.0.0	$2^{24} - 2$	2^7
Class B	128 – 191	10XXXXXXXX	128.0.0.0-191.255.255.255	255.255.0.0	$2^{16} - 2$	2^{14}
Class C	192 – 223	110XXXXXX	192.0.0.0-223.255.255.255	255.255.255.0	$2^8 - 2$	2^{21}
Class D (Multicast)	224 – 239	1110XXXXX	224.0.0.0-239.255.255.255			
Class E (Experimental)	240 – 255	1111XXXXX	240.0.0.0-255.255.255.255			

IPv4 Addresses (3 of 5)

- Class A, B, and C licensed IP addresses are available for use on the Internet
 - These are called **public IP addresses**
- A company can use **private IP addresses** on its private networks
- The IANA recommends the following IP addresses be used for private networks:
 - 10.0.0.0 through 10.255.255.255
 - 172.16.0.0 through 172.31.255.255
 - 192.168.0.0 through 192.168.255.255
- **Classless addressing** allows the dividing line between network and host portions to fall anywhere along the string of binary bits in an IP address
- **CIDR (Classless Interdomain Routing) notation** takes the network ID or a host's IP address and follows it with a forward slash (/) followed by the number of bits that are used for the network ID

IPv4 Addresses (4 of 5)

- **Network Address Translation (NAT)** is a technique designed to conserve public IP addresses needed by a network.
- Address translation is a process where a gateway device substitutes the private IP addresses with its own public address when the private devices connected to the gateway devices need access to other networks or Internet.
 - It allows a single public IP address to be used by many devices to access external services, without affecting the internal security.
- Gateway keeps track of which local host to receive response from external servers using Port Address Translation (PAT).
- PAT is the process of assigning a separate TCP port to each session between a local host and Internet host.
- Two variations of NAT to be aware of:
 - **SNAT (Source Network Address Translation)** - the gateway assigns the same public IP address to a host each time it makes a request to access the Internet
 - **DNAT (Dynamic Network Address Translation)** - the gateway has a pool of public address that it is free to assign to a local host when it makes a request to access the Internet. When a message sent to the public IP address reaches the router managing DNAT, the destination IP address is changed to the private IP address of the host inside the network.

IPv4 Addresses (5 of 5)

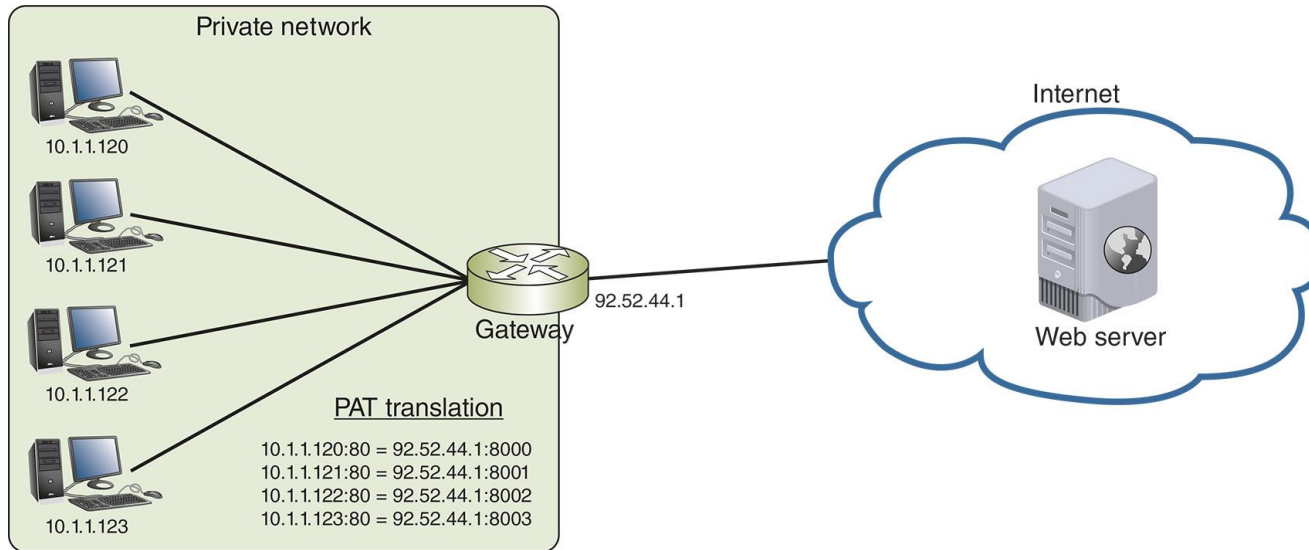


Figure 3-12 PAT (Port Address Translation)

Figure 3-12 PAT (Port Address Translation)

IPv6 Addresses (1 of 2)

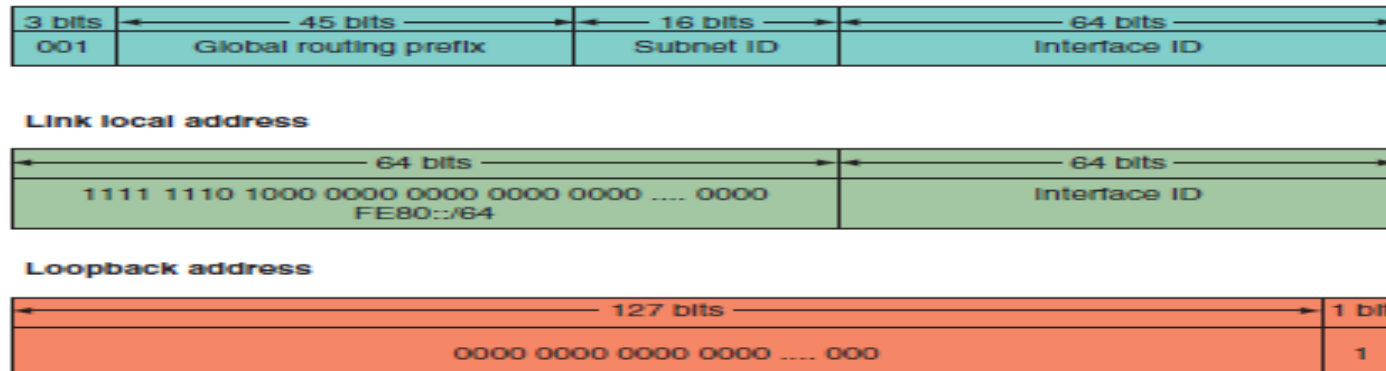
- An IPv6 address has 128 bits written as eight blocks of hexadecimal numbers separated by colons
 - Ex: 2001:0000:0B80:0000:0000:00D3:9C5A:00CC
 - Each block is 16 bits
 - Leading zeros in a four-character hex block can be eliminated
 - If blocks contain all zeroes, they can be written as double colons (::), only one set of double colons is used in an IP address
 - Therefore, above example can be written two ways:
 - 2001::B80:0000:0000:D3:9C5A:CC
 - 2001:0000:B80::D3:9C5A:CC (this is the preferred method because it contains the fewest zeroes)
- ▶ Any remaining block of four zeroes can be written as a single zero.
2001:0:B80::D3:9C5A:CC

IPv6 Addresses (2 of 2)

- IPv6 terminology:
 - A **link** (sometimes called local link) is any LAN bounded by routers.
 - **Neighbors** are two or more nodes on the same link.
 - **Dual stacked** is when a network is configured to use both IPv4 and IPv6.
 - **Tunneling** is a method used by IPv6 to transport IPv6 packets through or over an IPv4 network.
 - **NAT64** is a translation process that support compatibility between IPV4 and IPV6 devices.
 - **Interface ID** is the last 64 bits or four blocks of an IPv6 address that identify the interface.

Types of IPv6 Addresses (1 of 3)

- **Unicast address** - specifies a single node on a network
 - **Global address** can be routed on the internet and it is similar to the public IPv4 addresses.
 - **Link local address** can be used for communicating with nodes in the same link
 - **Loopback address** can be used to test that an interface and supporting protocol stack are functioning properly



- **Multicast address** – delivers packets to all nodes on a network
- **Anycast address** - can identify multiple destinations, with packets delivered to the closest destination

Types of IPv6 Addresses (2 of 3)

```
PS C:\Users\jillw> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VirtualBox Host-Only Network:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::2985:d1d8:60d2:4b2a%11
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 4:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::1c33:fed2:faf0:ece%14
    IPv4 Address. . . . . : 192.168.2.137
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.2.1

PS C:\Users\jillw>
```

IPv6 address assigned to virtual interface

IPv4 address assigned to virtual interface

IPv6 address assigned to physical Wi-Fi interface

IPv4 address assigned to physical Wi-Fi interface

Figure 3-18 The `ipconfig` command shows IPv4 and IPv6 addresses assigned to this computer

Figure 3-18 The `ipconfig` command shows IPv4 and IPv6 addresses assigned to this computer

Types of IPv6 Addresses (3 of 3)

- IPv6 autoconfiguration
 - IPv6 addressing is designed so that a computer can autoconfigure its own link local IP address
 - This process is called **SLAAC (stateless address autoconfiguration)**
- Step 1 - The computer creates its IPv6 address
 - It uses FE80::/64 as the first 64 bits (called prefix)
 - The last 64 bits are generated from the network adapter's MAC address
- Step 2 - The computer checks to make sure its IP address is unique on the network
- Step 3 - The computer asks if a router on the network can provide configuration information
 - This message is called an **RS (router solicitation)** message

Knowledge Check Activity 3-1

Which of the following IPv4 addresses is a public IP address?

- a. 10.0.2.14
- b. 172.16.156.254
- c. 192.168.72.73
- d. 64.233.177.189

Knowledge Check Activity 3-1: Answer

Which of the following IPv4 addresses is a public IP address?

Answer: d. 64.233.177.189

IP addresses within the ranges of 10.0.0.0 through 10.255.255.255, 172.16.0.0 through 172.31.255.255, and 192.168.0.0 through 192.168.255.255 are RFC1918, or private, IP addresses. The address 64.233.177.189 is a public IP address.

Ports and Sockets (1 of 2)

- A port is a number assigned to a **process**, such as an application or a service, that can receive data.
 - Port numbers ensure data is transmitted to the correct process among multiple processes running on a single device.
 - They range from 0 – 65535.
- A socket consists of host's IP address and the port number of an application running on the host. It manages an active data flow or listening state between a client and a server.
 - A colon separates the two values
 - Example - 10.43.3.87:23
- Port numbers are divided into three types:
 - Well-known ports - 0 to 1023
 - Registered ports - 1024 to 49151
 - Dynamic and private ports - 49152 to 65535

Ports and Sockets (2 of 2)

For example, the standard port for the SSH (Secure Shell) service is TCP 22. If a host has an IP address of 10.43.3.87, the socket for SSH running on that host is 10.43.3.87:22.

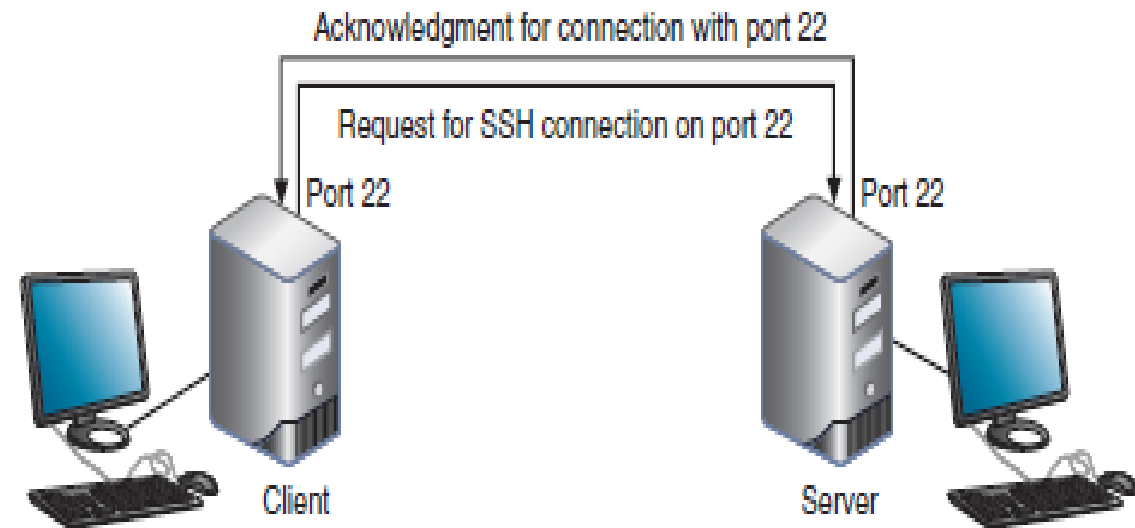


Figure 3-19 A virtual connection for the Telnet service

Domain Names and DNS (1 of 2)

- Character-based names are easier to remember than numeric IP addresses
- A **URL (uniform resource locator)** is an addressing scheme that identifies where to find a particular resource on a network
- Last part of an FQDN is called the **top-level domain (TLD)**
- Domain names must be registered with an Internet naming authority that works on behalf of ICANN
 - ICANN restricts what type of hosts can be associated with .arpa, .mil, .int, .edu, and .gov
- Name resolution is the process of discovering the IP address of a host when you know the FQDN

Domain Names and DNS (2 of 2)

- DNS is an Application layer client-server system of computers and databases made up of these elements:
 - **Namespace** - the entire collection of computer names and their associated IP addresses stored in databases on DNS name servers around the globe
 - **Name servers** - hold databases, which are organized in a hierarchical structure
 - **Resolvers** - a DNS client that requests information from DNS name servers

Namespace Databases

- Each organization that provides host services is responsible for providing and maintaining its own DNS authoritative servers for public access
 - An **authoritative name server** is the authority on computer names and their IP addresses for computers in their domains
- The domains that the organization is responsible for managing are called a **DNS zone**

Name Servers (1 of 4)

- Four common types of DNS servers:
 - **Primary DNS server** – the authoritative name server for the organization
 - Holds the authoritative DNS database for the organization's zones
 - **Secondary DNS server** – backup authoritative name server for the organization
 - **Caching DNS server** – accesses the public DNS data and caches the DNS information it collects
 - **Forwarding DNS server** – receives queries from local clients but doesn't work to resolve the queries
- Any of these DNS server types can co-exist on the same machine
- DNS name servers are organized in a hierarchical structure
- At the root level, 13 clusters of **root DNS servers** hold information used to locate top-level domain (TLD) servers

Name Servers (2 of 4)

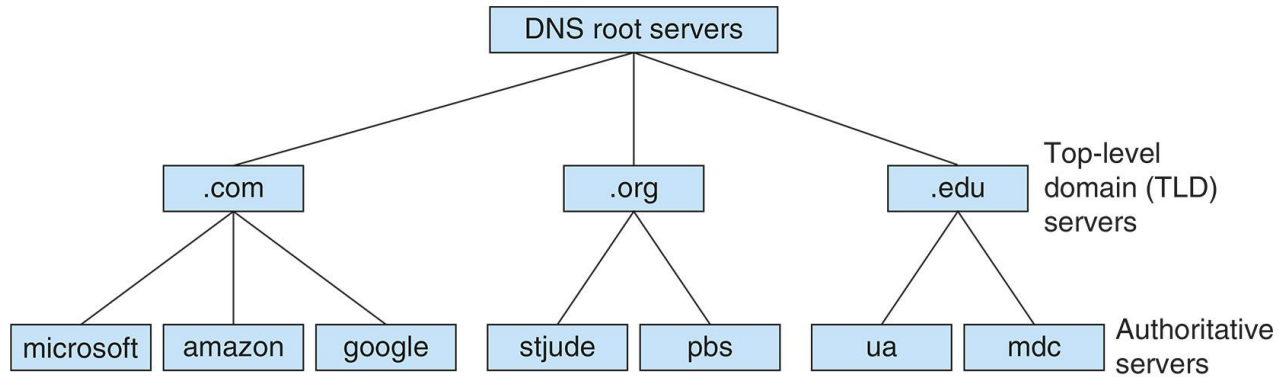


Figure 3-21 Hierarchy of name servers

Figure 3-21 Hierarchy of name servers

Name Servers (3 of 4)

Step 1 The resolver on the client computer first searches its DNS cache, a database stored on the local computer, for the match. If it can't find the information there, the resolver sends a DNS message or query to its local DNS server. In this example, let's assume this caching server doesn't yet know the IP address of the `www.mdc.edu` host.

Steps 2 and 3 The local name server queries a root server with the request. The root server responds to the local name server with a list of IP addresses of TLD name servers responsible for the `.edu` suffix.

Steps 4 and 5 The local name server makes the same request to one of the TLD name servers responsible for the `.edu` suffix. The TLD name server responds with the IP address of the `mdc.edu` authoritative server.

Steps 6 and 7 The local name server makes the request to the authoritative name server at Miami Dade College, which responds to the Cengage name server with the IP address of the `www.mdc.edu` host.

Step 8 The local name server responds to the client resolver with the requested IP address. Both the Cengage name server and the Cengage client computer store the information in their DNS caches, and, therefore, don't need to ask again until that information expires.

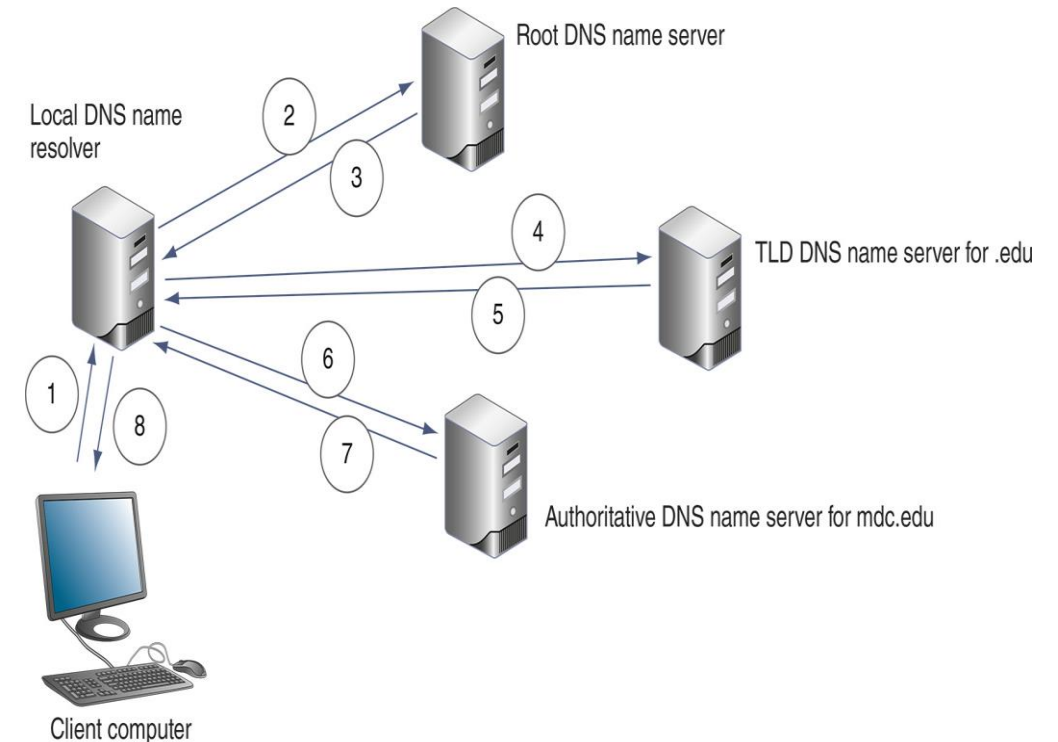


Figure 3-22 Queries for name resolution of `www.mdc.edu`

Name Servers (4 of 4)

- Ways the resolution process can get more complex:
 - A caching server typically is not the same machine as the authoritative server
 - The caching server exists only to resolve names for its own local clients
 - Name servers within a company might not have access to root servers
 - A TLD name server might be aware of an intermediate name server rather than the authoritative name server
- Two types of DNS requests:
 - **Recursive lookup** – a query that demands a resolution or the answer “It can’t be found”
 - **Iterative lookup** – a query where the local server issues queries to other servers
 - Other servers only provide information if they have it
 - Do not demand a resolution

Resource Records in a DNS Database

- Several types of records, called resource records are kept in a DNS database:
 - **SOA (start of authority) record** – gives information about the zone
 - **A (address) record** – stores the name-to-address mapping for a host
 - **AAAA (address) record** – holds the name-to-address mapping, the IP address is an IPv6 type IP address
 - **CNAME (canonical Name) record** – holds alternative names for a host
 - **PTR (pointer) record** – used for reverse lookups
 - **NS (name Server) record** – indicates the authoritative name server for a domain
 - **MX (mail exchanger) record** – identifies a mail server and is used for email traffic
 - **SRV (service) record** – identifies the hostname and port of a computer that hosts a specific network services besides email
 - **TXT (text) record** – holds any type of free-form text

DNS Server Software

- BIND (Berkeley Internet Name Domain) is the most popular DNS server software
 - Open source - the term for software whose code is publicly available for use and modification
- Microsoft DNS Server is a built-in DNS service in the Windows Server OS
- Windows Server is capable of split-brain or split-horizon deployment, which is used to handle internal clients and external clients

Troubleshooting Address Problems

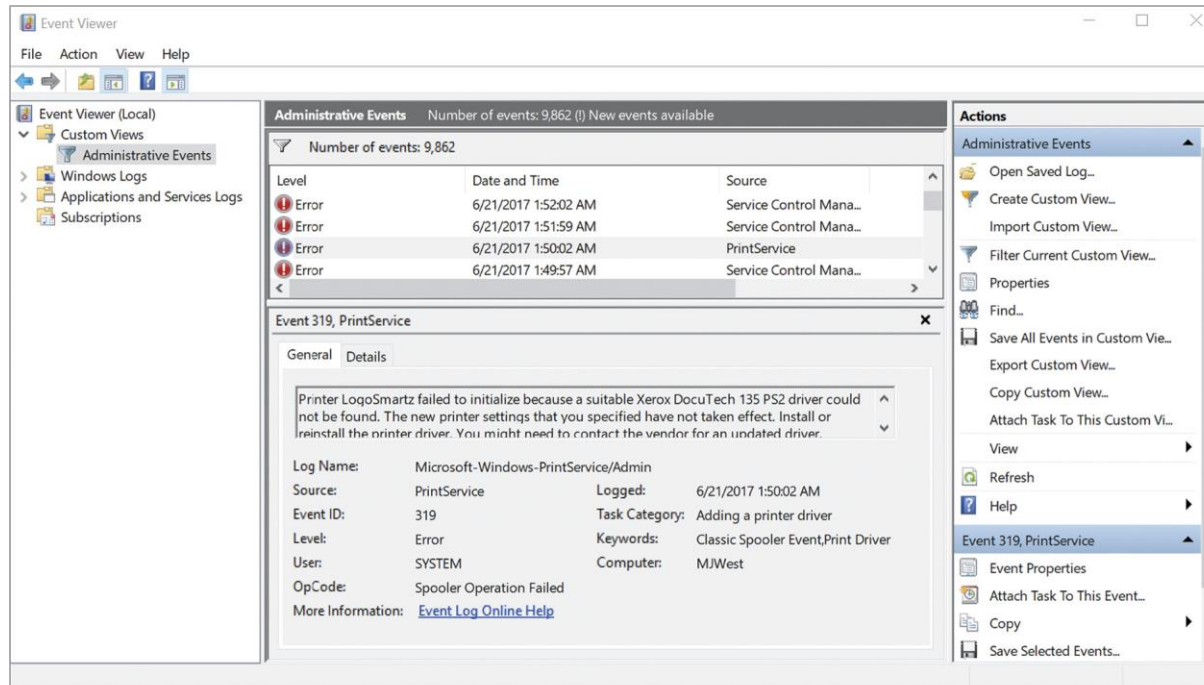


Figure 3-24 Event Viewer provided the diagnosis of a printer problem and recommended steps to fix the problem

Figure 3-24 Event Viewer provided the diagnosis of a printer problem and recommended steps to fix the problem

Troubleshooting Tools (1 of 8)

- Command-line tools are a great resource to troubleshoot network problems
- **ping (Packet Internet Groper)** utility is used to verify that TCP/IP is:
 - Installed
 - Bound to the NIC
 - Configured correctly
 - Communicating with the network
- The ping utility sends out a signal called an echo request to another device (request for a response)
 - The other computer responds in the form of an echo reply
- **ICMP (Internet Control Message Protocol)** is the protocol used by the echo request/reply to carry error messages and information about the network

Troubleshooting Tools (2 of 8)

- IPv6 networks use a version of ICMP called ICMPv6
 - ping6 – on Linux computers running IPv6, use `ping6` to verify whether an IPv6 host is available
 - ping -6 – on Windows computers, use `ping` with the `-6` switch to verify connectivity on IPv6 networks
- For the `ping6` and `ping -6` commands to work over the Internet, you must have access to the IPv6 Internet

Troubleshooting Tools (3 of 8)

- The **ipconfig** command shows current TCP/IP addressing and domain name information on a Windows computer
 - Use `ipconfig/all` to see a more complete summary of TCP/IP addressing information

Troubleshooting Tools (4 of 8)

```
Windows PowerShell

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
Description . . . . . : 802.11n USB Wireless LAN Card
Physical Address. . . . . : 7C-DD-90-72-62-56
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::1c33:fed2:faf0:ece%14(Preferred)
IPv4 Address. . . . . : 192.168.2.137(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Thursday, November 12, 2020 8:51:57 PM
Lease Expires . . . . . : Friday, November 13, 2020 3:31:11 AM
Default Gateway . . . . . : 192.168.2.1
DHCP Server . . . . . : 192.168.2.1
DHCPv6 IAID . . . . . : 75292048
DHCPv6 Client DUID. . . . . : 00-01-00-01-25-88-B7-19-70-85-C2-29-77-BD
DNS Servers . . . . . : 1.1.1.1
                       1.0.0.1
NetBIOS over Tcpi. . . . . : Enabled

PS C:\Users\jillw>
```

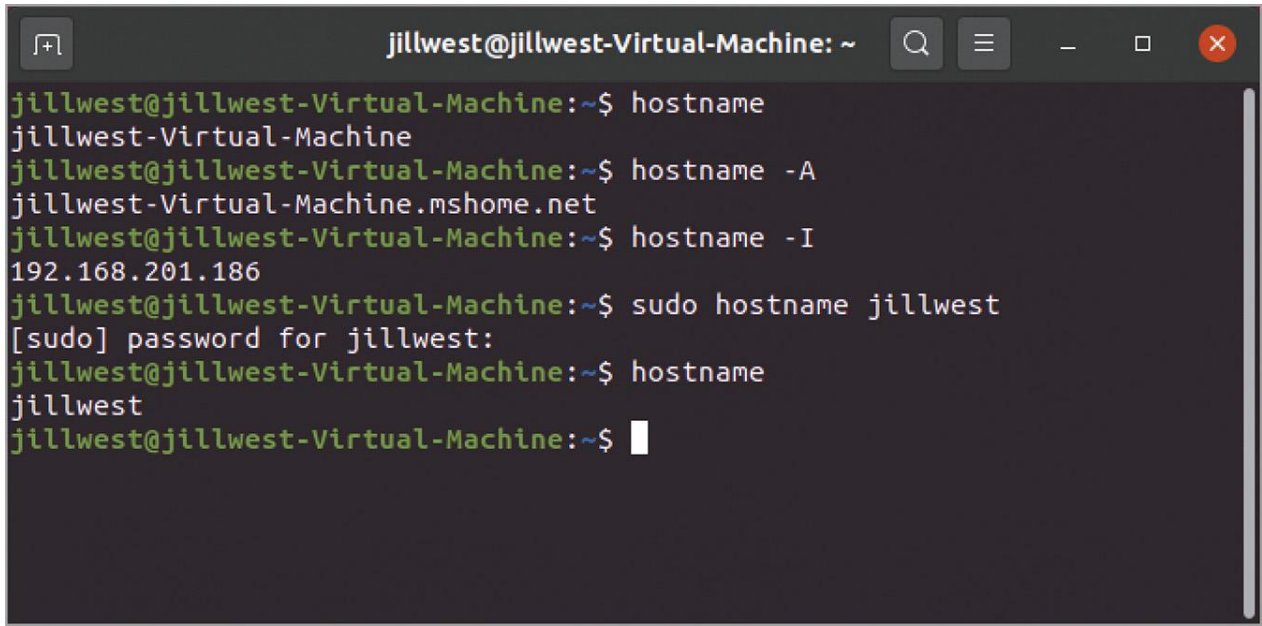
Figure 3-29 `ipconfig /all` gives more information than `ipconfig` by itself

Figure 3-29 `ipconfig /all` gives more information than `ipconfig` by itself

Troubleshooting Tools (5 of 8)

- Use the `ip` utility to view and manage TCP/IP settings
- The `ip` utility is only available on UNIX and Linux systems
- Any `ip` commands that change the state of a link require elevated privileges
 - This is accomplished by logging in as the root user or by temporarily elevating the current user's privileges with the `sudo` (superuser do) command
- `ifconfig` is a similar utility used to view and manage TCP/IP settings
- If your Linux or UNIX system provides a GUI
 - Open a shell prompt, then type `ifconfig`

Troubleshooting Tools (6 of 8)

A terminal window titled 'jillwest@jillwest-Virtual-Machine: ~' with standard window controls. The terminal shows the following commands and output:

```
jillwest@jillwest-Virtual-Machine:~$ hostname
jillwest-Virtual-Machine
jillwest@jillwest-Virtual-Machine:~$ hostname -A
jillwest-Virtual-Machine.mshome.net
jillwest@jillwest-Virtual-Machine:~$ hostname -I
192.168.201.186
jillwest@jillwest-Virtual-Machine:~$ sudo hostname jillwest
[sudo] password for jillwest:
jillwest@jillwest-Virtual-Machine:~$ hostname
jillwest
jillwest@jillwest-Virtual-Machine:~$
```

Source: The Linux Foundation

Figure 3-32 Use `hostname` to view or change a device's host name

Figure 3-32 Use `hostname` to view or change a device's host name

Troubleshooting Tools (7 of 8)

- The `nslookup` (name space lookup) utility allows you to query the DNS database from any computer on a network
 - To find the host name of a device by specifying its IP address, or vice versa
 - It is useful for verifying a host is configured correctly or for troubleshooting DNS resolution problems
- Reverse DNS lookup - to find the host name of a device whose IP address you know
 - `nslookup 69.23.208.74`
- The `nslookup` utility is available in two modes:
 - Interactive - to test multiple DNS servers at one time
 - Noninteractive - test a single DNS server
- You can change DNS servers from within interactive mode with the `server` subcommand and specifying the IP address of the new DNS server
- To exit `nslookup`'s interactive mode, enter `exit`

Troubleshooting Tools (8 of 8)

- The **dig (domain information groper)** utility is available on Linux and macOS
 - Provides more detailed information than nslookup and uses more reliable sources of information to output its results
- Use dig to query DNS nameservers for information about host addresses and other DNS records
- An **IP scanner** can be used to gather information about all devices connected to a network

Common Network Issues (1 of 2)

- Incorrect time
 - Check a domain computer's time source from a Command Prompt window by entering
`w32tm /query /source`
- DHCP Issues
 - If you are getting DHCP errors or if multiple clients are having trouble connecting to the network, try the following:
 - Check the settings on your DHCP server
 - Make sure the DHCP scope is large enough to account for the number of clients the network must support
 - Consider implementing a shorter lease time on larger networks

Common Network Issues (2 of 2)

- Network Connection Configuration Issues
 - Common configuration errors:
 - Incorrect IP address
 - Duplicate IP address
 - Incorrect subnet mask
 - Incorrect gateway
 - Incorrect DNS or DNS issues
 - When a computer is struggling to establish a network connection
 - Check its TCP/IP configuration settings
 - If the computer is not obtaining an IP address and related information from a DHCP server
 - Static settings might be using the wrong information
 - Try switching to DHCP

Knowledge Check Activity 3-2

What protocol does ping use?

- a. HTTP
- b. ICMP
- c. DHCP
- d. FTP

Knowledge Check Activity 3-2: Answer

What protocol does ping use?

Answer: b. ICMP

The protocol used by the ping echo request and echo reply is ICMP (Internet Control Message Protocol), a lightweight protocol used to carry error messages and information about a network.

Summary

Now that the lesson has ended, you should be able to:

- Work with MAC addresses
- Configure TCP/IP settings on a computer, including IP address, subnet mask, default gateway, and DNS servers
- Identify the ports of several common network protocols
- Describe domain names and the name resolution process
- Use command-line tools to troubleshoot common network problems